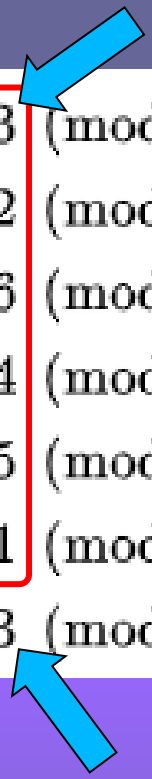
Diffie-Hellman Key Exchange

A number g is a **primitive root modulo n** if every number A coprime to n is congruent to a power of g modulo n.

That is,
g is a primitive root modulo n if
for every integer A coprime to n, there is an integer $k > 0$ such that $g^k \equiv A \pmod{n}$.

The number 3 is a primitive root modulo 7 because :

$$
\begin{aligned}
3^1 &= 3 = 3^0 \times 3 \equiv 1 \times 3 = 3 \equiv 3 \pmod 7 \\
3^2 &= 9 = 3^1 \times 3 \equiv 3 \times 3 = 9 \equiv 2 \pmod 7 \\
3^3 &= 27 = 3^2 \times 3 \equiv 2 \times 3 = 6 \equiv 6 \pmod 7 \\
3^4 &= 81 = 3^3 \times 3 \equiv 6 \times 3 = 18 \equiv 4 \pmod 7 \\
3^5 &= 243 = 3^4 \times 3 \equiv 4 \times 3 = 12 \equiv 5 \pmod 7 \\
3^6 &= 729 = 3^5 \times 3 \equiv 5 \times 3 = 15 \equiv 1 \pmod 7 \\
3^7 &= 2187 = 3^6 \times 3 \equiv 1 \times 3 = 3 \equiv 3 \pmod 7
\end{aligned}
$$



Here we see that the period of 3^k modulo 7 is 6.

Diffie-Hellman Setup

- All users agree on global parameters:
  1. large prime integer or polynomial q
  2. a being a primitive root mod q

- Each user (eg. A) generates their key
  1. chooses a secret key (number): $x_A < q$
  2. compute their **public key**: $y_A = a^{x_A} \bmod q$

- Each user makes public that key y_A

Diffie-Hellman Example

- Users Alice & Bob who wish to swap keys:
- Agree on prime $q=353$ and $a=3$
- Select random secret keys:
  - A chooses $x_A=97$, B chooses $x_B=233$
- Compute respective public keys:
  - $y_A=3^{97} \mod 353 = 40$ (Alice)
  - $y_B=3^{233} \mod 353 = 248$ (Bob)
- Compute shared session key as:
  - $K_{AB}= y_B^{x_A} \mod 353 = 248^{97} = 160$ (Alice)
  - $K_{AB}= y_A^{x_B} \mod 353 = 40^{233} = 160$ (Bob)

Diffie-Hellman Example

1. Alice and Bob publicly agree to use a modulus $p = 23$ and base $g = 5$ (which is a primitive root modulo 23).

2. Alice chooses a secret integer $\boldsymbol{a} = 4$, then sends Bob $A = g^{\boldsymbol{a}} \bmod p = 5^{\boldsymbol{4}} \bmod 23 = 4$

3. Bob chooses a secret integer $\boldsymbol{b} = 3$, then sends Alice $B = g^{\boldsymbol{b}} \bmod p = 5^{\boldsymbol{3}} \bmod 23 = 10$

4. Alice computes $\boldsymbol{s} = B^{\boldsymbol{a}} \bmod p = 10^{\boldsymbol{4}} \bmod 23 = 18$
5. Bob computes $\boldsymbol{s} = A^{\boldsymbol{b}} \bmod p = 4^{\boldsymbol{3}} \bmod 23 = 18$

6. Alice and Bob now share a secret (the number 18).

Alice		Bob		Eve	
Known	Unknown	Known	Unknown	Known	Unknown
$p = 23$		$p = 23$		$p = 23$	
$g = 5$		$g = 5$		$g = 5$	
$\mathbf{a = 6}$	$\mathbf{b}$	$\mathbf{b = 15}$	$\mathbf{a}$		$\mathbf{a}$, $\mathbf{b}$
$A = 5^{a} \bmod 23$		$B = 5^{b} \bmod 23$			
$A = 5^{6} \bmod 23 = 8$		$B = 5^{15} \bmod 23 = 19$			
$\mathbf{B = 19}$		$\mathbf{A = 8}$		$A = 8, B = 19$	
$\mathbf{s} = B^{a} \bmod 23$		$\mathbf{s} = A^{b} \bmod 23$			
$\mathbf{s} = 19^{6} \bmod 23 = \mathbf{2}$		$\mathbf{s} = 8^{15} \bmod 23 = \mathbf{2}$			$\mathbf{s}$

Now **s** is the shared secret key and it is known to both Alice and Bob, but *not* to Eve.

Note: It should be difficult for Alice to solve for Bob's private key or for Bob to solve for Alice's private key.

ALICE	BOB
Public Keys available = P, G	Public Keys available = P, G
Private Key Selected = a	Private Key Selected = b
Key generated = $x = G^a mod P$	Key generated = $y = G^b mod P$
Exchange of generated keys takes place	
Key received = y	key received = x
Generated Secret Key = $k_a = y^a mod P$	Generated Secret Key = $k_b = x^b mod P$
Algebraically it can be shown that $k_a = k_b$	
Users now have a symmetric secret key to encrypt	

Example

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Step 1: Alice and Bob get public numbers P = 23, G = 9

Step 2: Alice selected a private key a = 4 and
        Bob selected a private key b = 3

Step 3: Alice and Bob compute public values
Alice:    x =(9^4 mod 23) = (6561 mod 23) = 6
        Bob:    y = (9^3 mod 23) = (729 mod 23)  = 16

Step 4: Alice and Bob exchange public numbers

Step 5: Alice receives public key y =16 and
        Bob receives public key x = 6

Step 6: Alice and Bob compute symmetric keys
        Alice:  ka = y^a mod p = 65536 mod 23 = 9
        Bob:    kb = x^b mod p = 216 mod 23 = 9

Step 7: 9 is the shared secret.
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